Bracketing a root by bisection

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Given information. For exercise 1: seek a zero of $f(x) = x^2 - 2$ on $[1, 2]$. After two halvings, the retained interval is $[1.25, 1.5]$.

Exercise 1. Carry out a third step of the example.

Exercise 2. Starting with width 1, how many halvings guarantee width at most $\frac{1}{1000}$?

Exercise 3. Why does the sign change of $f(x) = \frac{1}{x}$ between -1 and 1 not prove the existence of a zero?

Exercise 4. A root lies in $[1.4, 1.5]$. What maximum error is guaranteed by midpoint 1.45?